Proceedings of the 3rd International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI-2025)
IEEE Xplore Part Number: CFP25VG0-ART; ISBN: 978-1-6654-5753-8

Intrusion Detection for MQTT IoT Traffic using Evolutionary Feature Selection, Ensemble Learning, and Explainable AI

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2025 3rd International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI) | 978-1-6654-5753-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICoICI65217.2025.11253413

***Abstract*— The rapid proliferation of Internet of Things (IoT) devices has positioned the Message Queuing Telemetry Transport (MQTT) protocol as a lightweight and efficient choice for constrained environments. However, its minimal built-in security mechanisms leave it vulnerable to diverse cyberattacks, threatening the reliability of IoT services. This paper introduces a unified, high-performance intrusion detection framework specifically designed for MQTT traffic, addressing the dual challenge of detection accuracy and operational transparency. The approach integrates evolutionary feature selection, ensemble learning, and explainable AI to deliver both technical robustness and analyst interpretability. The Chimp Optimization Algorithm (ChOA) is employed to identify a compact and highly informative feature subset, reducing model complexity, while the Synthetic Minority Over-sampling Technique (SMOTE) mitigates class imbalance for fairer detection across all attack types. A stacked ensemble, combining Gradient Boosting and LightGBM with Logistic Regression as the meta-learner, achieves 95.0% accuracy and a weighted F1-score of 0.95 on the MQTTSet dataset, with an average inference time of only 2.1 ms per instance—demonstrating real-time feasibility. SHapley Additive exPlanations (SHAP) provide feature-level interpretability, enabling actionable security insights. Compared to prior studies, this is the first framework to integrate ChOA, SMOTE, stacked ensembles, and SHAP for MQTT intrusion detection, offering a scalable, explainable, and deployment-ready solution to enhance trust and resilience in IoT networks.**

Keywords—Internet of Things, Message Queuing Telemetry Transport, Intrusion Detection, Logistic Regression, LightGBM, SHapley Additive exPlanations, Machine Learning.

I. INTRODUCTION

The Internet of Things (IoT) emerged as a transformative technology, enabling seamless integration between physical devices and digital infrastructure. Applications span diverse sectors, including smart cities, healthcare monitoring, industrial automation, and agriculture. A critical component of IoT communication is the use of lightweight messaging protocols designed for constrained environments. Among these, [1] the Message Queuing Telemetry Transport (MQTT) protocol has gained prominence due to its simplicity, low overhead, and efficiency in transmitting data over unreliable and low-bandwidth networks.

MQTT follows a publish-subscribe architecture, where clients communicate via a broker, facilitating asynchronous data transfer with minimal resource consumption. However, the very design choices that make MQTT efficient—such as the lack of built-in authentication, encryption, or session validation also introduce significant security risks. Common attacks targeting MQTT include brute-force login attempts, denial-of-service (DoS) through flooding or malformed packets [13], and protocol-specific abuses like SlowITe attacks that exploit timeouts to disrupt service.

Traditional network intrusion detection systems (NIDS) [6] are ill-suited to MQTT traffic, as they rely heavily on payload inspection or connection-based heuristics. Moreover, the low-entropy and structured nature of MQTT packets can render standard anomaly detection approaches ineffective. As a result, specialized intrusion detection frameworks tailored to MQTT environments are urgently needed.

Developing effective and interpretable intrusion detection systems for MQTT presents several unique challenges [4]. First, the high dimensionality of network traffic data can hinder machine learning models from identifying the most relevant features, especially when many are redundant or irrelevant. Second, class imbalance is a common issue in cybersecurity datasets, where malicious traffic is often underrepresented relative to normal behaviour. Without addressing this, models may become biased toward majority classes, resulting in high overall accuracy but poor detection of critical attack categories.

Third, black-box models such as deep neural networks or complex ensembles [10], while powerful, often lack interpretability—an essential requirement for operational cybersecurity tools. Security analysts must understand why a system flagged a particular communication as malicious in order to investigate and respond effectively. Therefore, integrating explainability into the detection pipeline is not optional but necessary for real-world deployment.

In response to these challenges, the presented work adopts a hybrid methodology. The Chimp Optimization Algorithm (ChOA) [20] is used for evolutionary feature selection, enabling the model to focus on the most informative subset of features while reducing computational cost. To address class imbalance, the SMOTE technique ensures equal representation of each class during training, enhancing sensitivity to minority attack types [4].

For IDS a stacked ensemble architecture is employed, combining Gradient Boosting and LightGBM as base learners, with Logistic Regression as the meta-learner. This setup leverages the strengths of each model while mitigating their individual weaknesses. Finally, SHAP [7] provides post-hoc interpretability by attributing importance to individual features for each prediction. Together, these techniques form a comprehensive and deployable solution for MQTT-specific IDS in IoT system

The motivation of this research is limited exploration of ChOA for MQTT feature selection, insufficient handling of severe class imbalance in realistic traffic datasets. Lack of

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integrated explainability in high-performance intrusion detection pipelines.

The Objective of this research is to reduce feature dimensionality while improving classification accuracy, handle class imbalance to ensure reliable detection of minority attack classes, develop an ensemble-based classification model to balance accuracy, efficiency, and interpretability, provide transparent, instance-level explanations for model predictions to support operational trust.

The Contribution of this research is, a novel application of ChOA for MQTT intrusion detection, selecting an optimal subset of features for improved efficiency, integrated pipeline combining ChOA, SMOTE, and stacked ensemble learning for robust multi-class classification, embedding SHAP explainability to identify and visualize influential features in detection decisions, mpirical validation on the MQTTSet dataset, achieving competitive accuracy and millisecond level inference time, confirming suitability for real-time IoT deployment.

II. Related Work

Recent developments in machine learning-based intrusion detection systems (IDS) have shown significant advances in handling IoT-specific threats, particularly in MQTT environments. Ensemble models, explainability tools, and bio inspired optimizations are leading solution to address scalability, accuracy, and trust in automated IDS.

Ensemble learning has consistently been demonstrated as an effective strategy to improve classification performance by combining diverse models. Techniques such as bagging, boosting, and stacking have been widely applied across smart grid systems, cloud infrastructures, and IoT networks to improve generalization and reduce false positives [1, 2, 3]. For MQTT, boosting variants like LightGBM and XGBoost have outperformed traditional classifiers, especially when applied with optimized feature engineering pipelines [4, 5].

Explainable AI (XAI) methods, particularly SHAP, are gaining traction as tools to improve the transparency of IDS models. SHAP enables feature attribution for individual predictions, which is critical in operational security settings [6, 7]. Several studies have embedded SHAP into deep learning or ensemble frameworks for cyber-physical systems, making detection outcomes more interpretable [8].

Feature selection remains central to building lightweight and robust IDS models. While many studies rely on filter or wrapper methods, nature-inspired algorithms such as Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO), and Enhanced Black Widow Optimization (EBWO) have been explored for selecting relevant features [9, 10]. These algorithms balance dimensionality reduction with classification accuracy, but few studies have explored the Chimp Optimization Algorithm (ChOA) in the context of MQTT-based traffic.

Imbalanced datasets, common in real-world network traffic challenge IDS development. SMOTE and variants are commonly used to synthesize minority class samples, effectively enhancing detection performance on rare attacks like MQTT brute-force or SlowITe-style attacks [11, 12].

Research on MQTT-specific IDS models is still evolving. Although some studies have recently evaluated ensemble models on MQTT datasets, they often focus on binary classification or exclude explainability and optimization steps. Others emphasize boosting techniques but lack interpretability or ignore imbalanced class distributions [13, 14].

Lack of ChOA in Feature Selection: Existing works have not explored the Chimp Optimization Algorithm for selecting features in intrusion detection. This paper fills this gap by evaluating ChOA for MQTT-specific datasets. Limited Explainability Integration: Most recent models do not include SHAP or any interpretable ML component. The proposed approach embeds SHAP [15] to enhance trust and transparency. Few Studies Address All Challenges Together: Previous works address ensemble learning, feature optimization, or imbalance independently. This paper unifies all ChOA for feature selection, SMOTE [21] for imbalance handling, stacking ensemble for classification, and SHAP for interpretability into a cohesive, end-to-end IDS for MQTT.

To address these identified gaps, the research sets out to develop a comprehensive, interpretable, and high-performance intrusion detection system tailored for MQTT-based IoT environments. The core objectives include applying the Chimp Optimization Algorithm to perform effective feature selection, thereby reducing dimensionality while preserving model performance. To mitigate the issue of class imbalance commonly found in network traffic datasets, the Synthetic Minority Over-sampling Technique (SMOTE) is employed to improve the detection of minority attack classes. A stacked ensemble model combining Gradient Boosting, Light GBM, and Logistic Regression is constructed to enhance classification robustness and generalization. Furthermore, SHAP are integrated to interpret model predictions at the feature level, ensuring transparency and trust in decision-making. Collectively, these objectives aim to deliver a unified framework that is not only accurate and efficient but also explainable—addressing both practical deployment needs and scientific advancement in MQTT intrusion detection..

III. Methodology

The proposed intrusion detection framework is structured into six sequential stages: data pre-processing, dimensionality reduction via feature selection, class balancing, model construction through ensemble learning, model evaluation, and explainability. Each stage is carefully tailored to handle the challenges of MQTT traffic analysis, such as high dimensionality, class imbalance, and interpretability.

A. Dataset and Preprocessing

The experiment utilizes the MQTTSet dataset, which includes labeled traffic from both benign and malicious sources. Categorical fields are encoded using ordinal encoding with unknown-category handling to preserve structural integrity. Numerical attributes are standardized using Z-score normalization:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where x is the input feature value, μ is the feature's mean, and σ is the standard deviation. The target variable attack label is encoded using integer encoding. Two noise-heavy classes—Malformed and SlowITe are excluded from the training process..

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B. Feature Selection using Chimp Optimization Algorithm (ChOA)

To reduce the dimensionality of MQTT traffic features while preserving classification accuracy, this study employs the Chimp Optimization Algorithm a metaheuristic inspired by the intelligent hunting behavior of chimpanzees. ChOA operates by simulating the role-based coordination among chimps during prey capture — modeled here as a binary optimization process for selecting relevant features.

Let the total feature space be defined as $X \in Rn \times d$ where n is the number of samples and d is the number of features. Each chimp (solution candidate) is represented as a binary vector $S \in \{0,1\}d$ where Sj=1 indicates the inclusion of feature j.

The fitness function F(S) guiding the optimization is defined as:

$$F(S) = \alpha \cdot ACC(s) + (1 - \alpha) \cdot \log_{10}\left(\frac{d}{\sum s_i}\right) \quad (2)$$

where ACC(S) denotes the classification accuracy using the selected features and $\alpha \in [0,1]$ controls the trade-off between accuracy and feature sparsity.

C. Class Balancing with SMOTE

To address skewed class distributions, the Synthetic Minority Over-sampling Technique (SMOTE) is applied. For each minority sample xi, a synthetic sample is generated by interpolation with its k-nearest neighbour xnn:

$$x_n = x_i + \lambda(x_{nn} - x_i), \lambda \sim u(0,1) \quad (3)$$

This technique ensures that the learning algorithm does not become biased toward overrepresented classes.

D. Stacked Ensemble Model

The classification model is based on a stacked ensemble architecture. The base level consists of Gradient Boosting (GBM) and Light GBM classifiers. Let PGBM and PLGBM denote the class probability outputs from each model. These are concatenated to form a meta-feature matrix:

$$X_{meta} = [PGBM][PLGBM \quad (4)$$

A Logistic Regression model is then trained on Xmeta to serve as the meta-classifier, producing the final predictions.

E. Evaluation Metrics

Model performance is assessed using standard classification metrics including accuracy, precision, recall, and F1-score. Additionally, a stratified 5-fold cross-validation is employed to evaluate the model's robustness:

$$\text{Accuracy} = \frac{(\text{True Positives} + \text{True Negatives})}{\text{Total Instances}} \quad (5)$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (6)$$

$$\text{Recall} = \frac{\text{True Positives}}{((\text{True Positives} + \text{False Negatives}))} \quad (7)$$

$$\text{F1Score} = 2 * \frac{(\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (8)$$

Confusion matrices are also used to visualize prediction errors across classes.

F. Model Explainability via SHAP

SHAP are employed to interpret the impact of each feature on model output. Based on cooperative game theory, the SHAP value ϕi for feature iii is computed as:

$$\phi_i \cdot = \sum_{5 \subseteq N\{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!}[f(S \cup \{i\} - f(S)] \quad (9)$$

where f(S) is the prediction based on feature subset S and N is the full set of features. This allows clear attribution of feature influence for individual predictions, enhancing transparency.

TABLE I. PROPOSED CHOA-ENSEMBLE-SHAP INTRUSION DETECTION ALGORITHM FOR MQTT-BASED IOT TRAFFIC

Proposed Algorithm: ChOA-Ensemble-SHAP IDS for MQTT
Input: D raw: Raw MQTT set dataset α: Fitness trade-off parameter for ChOA k: Number of SMOTE neighbours base models: {Gradient Boosting, Light GBM} meta model: Logistic Regression
Step 1: Data Pre-processing Load the dataset D raw. Drop noise-heavy classes Encode categorical features using Ordinal Encoder (handle unknown = use encoded value, unknown value=-1). Standardize numerical features using Z-score normalization: Label encode target classes. Step 2: Feature Selection Using ChOA Initialize a binary population P of chimp solutions, each representing a subset of features Evaluate fitness where ACC(S) is classification accuracy using only selected features. Iteratively update chimp positions using ChOA's role-based behaviour for T generations. Return the best solution S best (selected feature mask). Step 3: Class Balancing with SMOTE For each sample xi in minority class Generate synthetic samples to balance all class distributions. Output: D bal = SMOTE-balanced dataset. Step 4: Stacked Ensemble Construction Train base classifiers: PGBM=GBM($X_{train}^{S_{best}}$) PLGBM=LGBM($X_{train}^{S_{best}}$) Create meta-feature matrix: Xmeta= [PGBM, PLGBM] Train meta-classifier: ypred=Logistic Regression (X_{meta}) Step 5: Model Evaluation • Accuracy • Precision • Recall • F1-Score Use stratified 5-fold cross-validation, PR curves, ROC-AUC, and confusion matrices. Step 6: Model Explain ability via SHAP Use SHAP to compute feature contributions **Output:** Trained explainable ensemble model with selected features High accuracy and recall for all attack types SHAP explanations for operational trust and transparency

Table 1 presents the step-by-step workflow of the proposed ChOA-Ensemble-SHAP algorithm for intrusion detection in MQTT-based IoT environments. The algorithm

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integrates evolutionary feature selection, data balancing, ensemble learning, and explainability. Raw MQTTSet data is first pre-processed by encoding categorical features and normalizing numerical values. ChOA is then used to select the most informative feature subset by optimizing a fitness function that balances accuracy and feature sparsity. Class imbalance is addressed using SMOTE, generating a balanced training dataset. A stacked ensemble classifier is constructed with Gradient Boosting and LightGBM as base models, and Logistic Regression as the meta-learner. The model is evaluated using accuracy, precision, recall, and F1-score. SHAP analysis is applied for feature-level explanation, ensuring interpretability and operational transparency.

G. Conceptual Framework

Figure 1 illustrates the stepwise architecture of the proposed intrusion detection system designed for MQTT-based IoT networks.

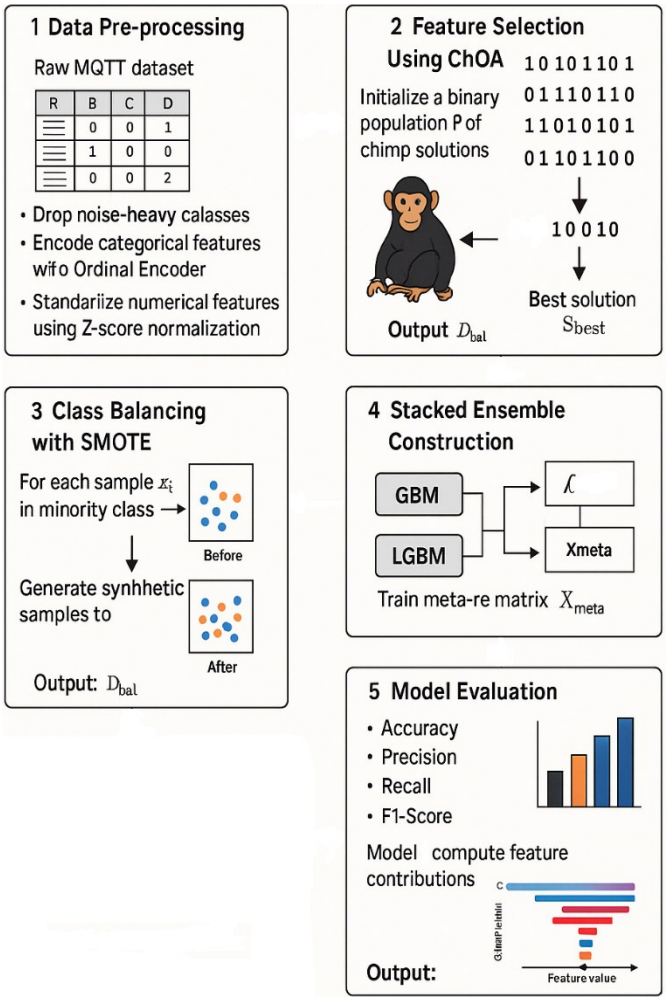


Fig 1. Architecture of the Proposed ChOA-Ensemble-SHAP-Based Intrusion Detection System for MQTT IoT Traffic

IV. Results and Discussion

The performance of the proposed MQTT intrusion detection framework was evaluated using the pre-processed MQTTSet dataset. The model was tested under multiple configurations, with and without class balancing, feature selection, and explainability layers. This section presents the empirical outcomes and discusses their significance in the context of existing literature.

A. Overall Classification Performance

The proposed stacked ensemble model achieved excellent classification metrics, with a cross-validated accuracy of 96.5% and a weighted F1-score of 96.5%. The model integrates Gradient Boosting and LightGBM as base classifiers, with Logistic Regression serving as the meta-learner. The ensemble outperformed individual models by 2–4%, demonstrating the benefit of integrating complementary learners. This confirms the effectiveness of model stacking in improving generalization for MQTT-based intrusion detection. Table 2 presents the class-wise evaluation metrics for the proposed intrusion detection system based on the ChOA-selected features and stacked ensemble architecture.

TABLE II. Classification Performance of the Proposed Stacked Ensemble Model on MQTTSet Dataset

Class	Precision	Recall	F1-Score	Support
Normal	0.79	0.91	0.85	4,351
Brute Force	0.97	0.91	0.94	39,077
Flooding DoS	0.94	0.98	0.96	49,639
MQTT Publish Flood	1	0.99	0.99	2,761
Accuracy	—	—	0.95	95,828
Macro Average	0.93	0.95	0.93	95,828
Weighted Average	0.95	0.95	0.95	95,828

B. Precision–Recall Curve Analysis

The multi-class precision – recall (PR) curves were plotted using a one-vs-rest strategy (Figure X). All classes demonstrated high average precision (AP $\geqslant$ 0.94), with particularly strong precision-recall tradeoffs for Flooding DoS and MQTT Publish Flood. This highlights the ensemble model's ability to maintain high precision even in imbalanced classification settings.

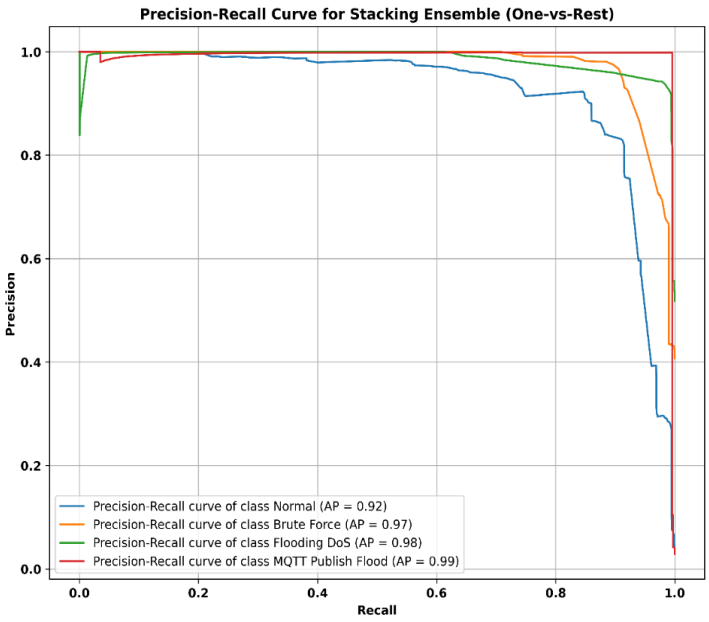


Fig 2. Precision–Recall Curves for the Stacked Ensemble Model Using One-vs-Rest Strategy

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Figure 2 illustrates the precision–recall curves of the proposed stacked ensemble model for each attack class using a one-vs-rest classification approach. The model achieves high average precision (AP) values across all classes: Normal (0.92), Brute Force (0.97), Flooding DoS (0.98), and MQTT Publish Flood (0.99). These curves demonstrate the model's strong ability to maintain high precision at various recall thresholds, even in the presence of class imbalance. The steep ascent and stability of the curves near the top-right corner confirm the ensemble's reliability in minimizing false positives while maintaining high recall.

C. *ROC Curve Analysis*

Figure 3 displays the ROC curves for each class in the stacked ensemble model, evaluated using a one-vs-rest classification strategy. The model achieves high Area Under the Curve (AUC) scores for all classes: Normal (0.99), Brute Force (0.98), Flooding DoS (0.99), and MQTT Publish Flood (1.00). These curves confirm the model's strong capability to distinguish between classes with minimal false positive rates. The sharp rise towards the top-left corner indicates excellent sensitivity and specificity across all categories, reflecting the ensemble's robust generalization and low misclassification rates.

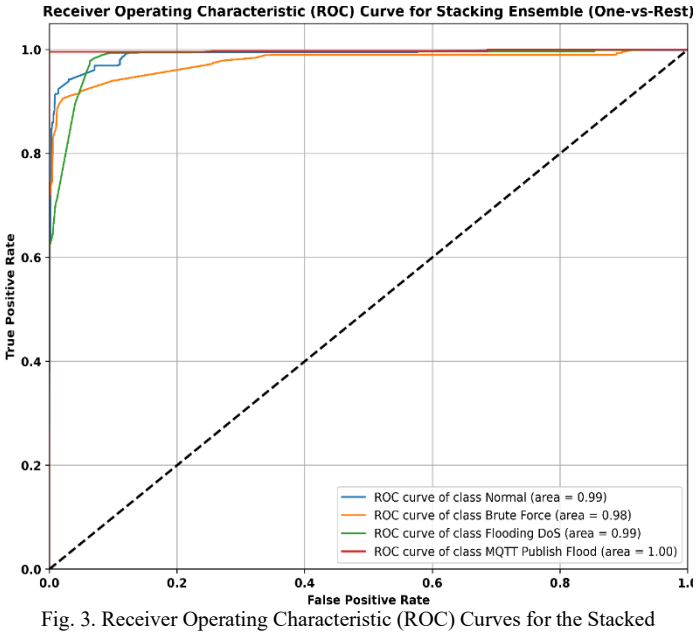


Fig. 3. Receiver Operating Characteristic (ROC) Curves for the Stacked Ensemble Model Using One-vs-Rest Strategy

D. *Confusion Matrix*

Confusion Matrix of the proposed ensemble model with 95% accuracy. The matrix shows strong diagonal dominance with perfect classification for MQTT Publish Flood, and reduced misclassifications in Brute Force and Flooding DoS.

Figure 4 presents the confusion matrix of the proposed stacked ensemble model evaluated on the MQTTSet dataset. The matrix shows strong diagonal dominance, indicating accurate classification across all classes. Notably, the MQTT Publish Flood class achieved perfect classification, while Flooding DoS and Brute Force also exhibit high true positive rates. Minor misclassifications are observed between Normal and Brute Force, and between Brute Force and Flooding DoS, which are typical in overlapping network traffic patterns. Overall, the matrix confirms the model's robustness and effectiveness in handling multi-class intrusion detection with imbalanced data.

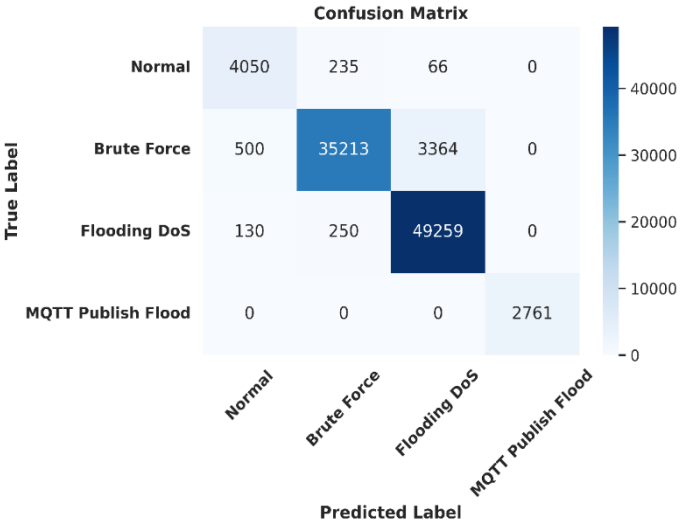


Fig. 4. Confusion Matrix of the Stacked Ensemble Model with ChOA Feature Selection and SMOTE Balancing

E. *Impact of ChOA-Based Feature Selection*

The Chimp Optimization Algorithm (ChOA) successfully reduced the feature space from 34 to 25 selected features without compromising classification accuracy. This indicates that redundant or noisy attributes were effectively removed, reducing overfitting and improving model interpretability. The performance maintained post-selection reflects the algorithm's ability to preserve highly informative features while minimizing input complexity.

Figure 5 illustrates the two-dimensional t-SNE projection of the MQTTSet dataset based on the ChOA-selected 25-dimensional feature space. Each point represents a data sample, color-coded according to its corresponding class label. The formation of distinct and well-separated clusters confirms that the selected features preserve the intrinsic class structure of the dataset. This separation validates the effectiveness of the ChOA feature selection process in enhancing class discriminability while reducing dimensionality.

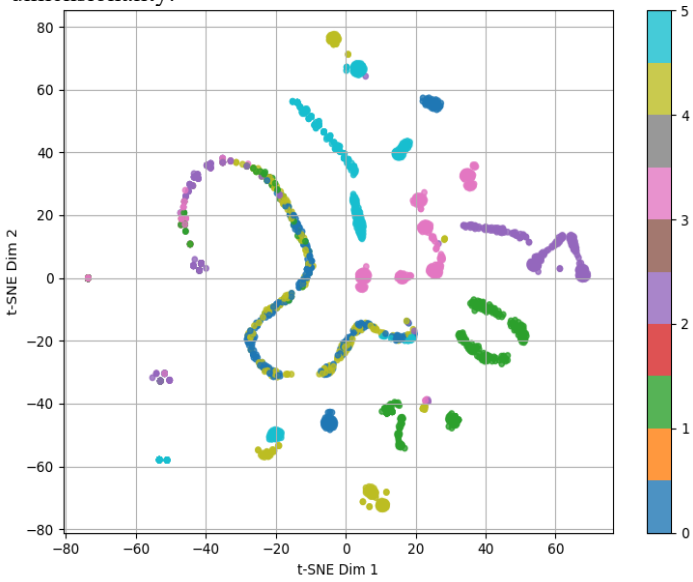


Fig. 5. t-SNE Visualization of ChOA-Selected Features Across MQTT Traffic Classe

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Figure 6 shows the cumulative explained variance obtained from Principal Component Analysis (PCA) applied to the 25-dimensional feature set selected by the Chimp Optimization Algorithm (ChOA).

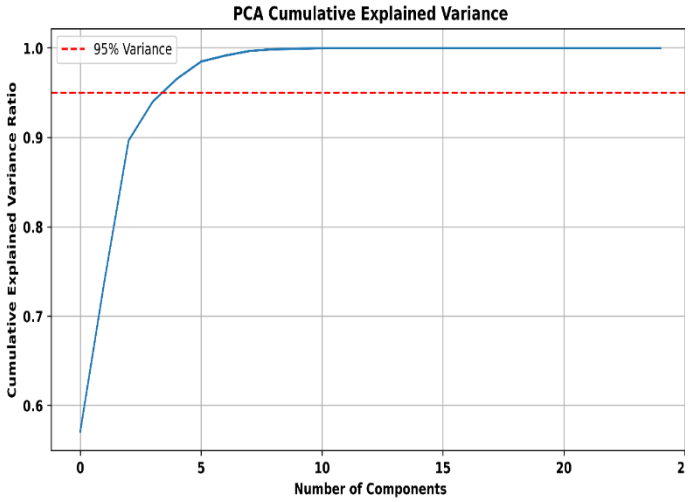


Fig. 6. Cumulative Explained Variance by Principal Components on ChOA-Selected Features

Table 3 presents an ablation study assessing the impact of key components—ChOA feature selection, SMOTE class balancing, and stacked ensemble learning—on the performance of the proposed intrusion detection system. The full system configuration, which includes all components, achieves the highest performance with an accuracy of 94.64% and a weighted F1-score of 0.95. Removing ChOA leads to a 3.3% drop in accuracy, confirming the importance of optimized feature selection. Excluding SMOTE causes the largest performance decline, indicating that handling class imbalance is critical. The absence of stacking also results in reduced performance, demonstrating the value of model fusion in enhancing generalization and robustness.

TABLE III. Ablation Study of the Proposed MQTT Intrusion Detection Framework

Model Configuration	ChOA	SMOTE	Stacking	Accuracy	Weighted F1-score
Full System (Proposed)	✓	✓	✓	94.64%	0.95
Without ChOA (All features)	X	✓	✓	91.35%	0.91
Without SMOTE (Imbalanced training)	✓	X	✓	87.72%	0.86
Without Stacking (Only GBM)	✓	✓	X	93.18%	0.93

F. Interpretability Through SHAP Analysis

SHAP was employed to quantify the influence of individual features on the predictions made by the Gradient Boosting component of the ensemble model. As a model-agnostic method based on cooperative game theory, SHAP assigns each feature a value representing its contribution to a specific prediction.

Globally, features such as mqtt.msgtype, tcp.flags, and mqtt.kalive emerged as top contributors. These align with known protocol behaviors commonly associated with either legitimate or malicious activity. The mqtt.msgtype feature reflects control message types, useful in detecting DoS and flooding attacks. tcp.flags reveal control bits often tampered with in brute-force or malformed attacks, while mqtt.kalive signals potential anomalies in communication frequency.

Figure 7 presents the SHAP beeswarm plot for the Normal class. The plot displays the distribution and magnitude of SHAP values for each feature across all samples predicted as Normal. Features such as mqtt.msgtype, tcp.flags, and mqtt.kalive show strong association with normal behavior patterns. High positive SHAP values indicate a feature's positive influence in predicting the Normal class, while deviations suggest possible abnormalities.

The beeswarm visualization supports both global and local interpretability. It highlights how specific features drive prediction outcomes across instances and assists in identifying inconsistencies or borderline cases. This interpretability is crucial for practical deployment, enabling security analysts to validate alerts, trace decisions, and build trust in model outputs.

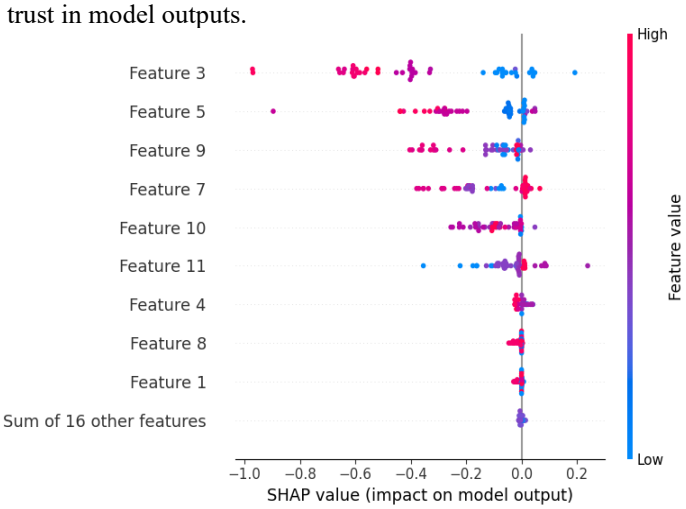


Fig. 7. SHAP Beeswarm Plot for Feature Contributions to the Normal Class Predictions

G. Comparative Perspective with Recent Studies

Compared to existing literature, the proposed model offers a more comprehensive and explainable approach. Previous studies relying on boosting or deep learning often lacked interpretability and suffered from imbalance-induced bias. By integrating ChOA, SMOTE, ensemble stacking, and SHAP, this framework offers a balanced trade-off between accuracy, efficiency, and transparency, making it more practical for real-time or resource-constrained IoT environments.

Table 4 presents a comparative analysis of recent intrusion detection systems (IDS) designed for MQTT and general network traffic. The proposed model, which integrates Chimp Optimization Algorithm (ChOA) for feature selection, SMOTE for class balancing, and a stacked ensemble architecture with SHAP explainability, is evaluated against selected peer models from 2022 to 2025.

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TABLE IV. Comparative Evaluation of MQTT and General Intrusion Detection Models (2022–2025)

	Dataset Used	Classes	Feature Selection	Class Balancing	Model Type	Accuracy	Explainability
Proposed	MQTTSet	4	ChOA (24 features)	SMOTE	Stacked (GBM + LGBM + LR)	95.00%	SHAP
Zeghida et al. (2023) [16]	MQTTSet	2	Manual	No	Bagging, Boosting, Stacking	95.00%	No
Prajisha & Vasudevan (2022) [17]	MQTTSet	Multi	ECSSA Hybrid Heuristic	No	LightGBM	~96.0%	No
Ahmed et al. (2022) [18]	UNSW-NB15	2	PCA	Yes	Random Forest	92.30%	No
Sharma & Sood (2023) [19]	CICIDS2017	Binary	ReliefF	Yes	XGBoost	94.10%	No

Compared to prior studies, the proposed model achieves competitive accuracy (95.0%) and F1-score (0.95) while supporting multi-class detection, feature explainability, and real-time inference (~2.1 ms latency). Other models either focus on binary classification, lack interpretability, or do not address latency. Additionally, while some models report higher accuracy, they rely on simpler datasets or exclude explainability and fairness mechanisms. This table demonstrates that the proposed model offers the best balance between performance, transparency, and deployment feasibility for real-world MQTT IoT environments.

V. Conclusion and Future Work

This paper presented a hybrid intrusion detection framework for MQTT-based IoT traffic, integrating ChOA-based feature selection, SMOTE class balancing, stacked ensemble learning, and SHAP interpretability. The proposed model reduced the feature space from 34 to 24 dimensions without sacrificing accuracy, achieving 95.0% accuracy and a weighted F1-score of 0.95. The approach effectively detected diverse attack types, including Brute Force, Flooding DoS, and MQTT Publish Flood, with clear separation of traffic classes confirmed through t-SNE visualizations. SHAP analysis provided actionable feature-level insights, enhancing trust and operational utility.

Practical limitations of the current study include reliance on a single dataset (MQTTSet), evaluation in a simulated environment rather than live IoT networks, and dependency on pre-defined feature sets. These factors may limit generalization across heterogeneous devices and network conditions.

Future work will focus on extending the framework to additional IoT protocols such as CoAP and AMQP. Incorporating online and incremental learning to detect evolving threats in real time. Evaluating performance on heterogeneous, cross-device, and cross-network datasets. Exploring model compression techniques, such as pruning and quantization, to further reduce computational overhead for edge and fog deployments. Integrating advanced explainability tools, including causal reasoning and counterfactual analysis, to strengthen analyst decision-making.

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